Customer segmentation with clustering

Analysis report

Author: Victoria Durrant

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Table of Contents

Introduction	3
Methods	4
Results and Discussion	5
Conclusion	12

Introduction

Understanding and serving customers are the two marketing objectives. This is achieved by customer-centric approaches deciding continual improvements to metrics including marketing efficiency and customer satisfaction.

Customer segmentation allows a business to group customers based on demographics, geographics, psychographics, behaviour, technographic and needs. This project aims to explore the data, perform customer segmentation with clustering models for insight. This will assist the e-commerce company in understanding and serving its customers better and improve marketing efforts.

Methods

The dataset used in this project consisted of 951,668 observations, representing customer ordered products using 20 features.

Data preprocessing was performed on the raw data to improve data quality and to ensure the dataset was suitable for customer segmentation. The dataset was examined for missing values, duplicate records, and data type assignments.

The resulting dataset was used for customer-level aggregation, feature selection and engineering (frequency, recency, customer lifetime value, Average_Unit_Cost and customer age).

EDA

EDA was performed to view the dataset's characteristics, this included feature distributions, identifying potential correlations and relationships, outliers, and descriptive statistics for central tendencies and summary.

Clustering with ML models

Standard scaling was applied to the selected features and used as the input model for unsupervised machine learning clustering models and dimension reduction.

K-Means clustering was applied, and the number of clusters were supported by the Elbow and Silhouette scores. Hierarchical clustering was also performed and a dendrogram created to allow visualisation of the structure, and formation of the clusters to support K-means.

Visualising customer segments and dimensionality reduction

Box plots, PCA and t-SNE were used to visualise the clusters and interpretability of customer behaviour.

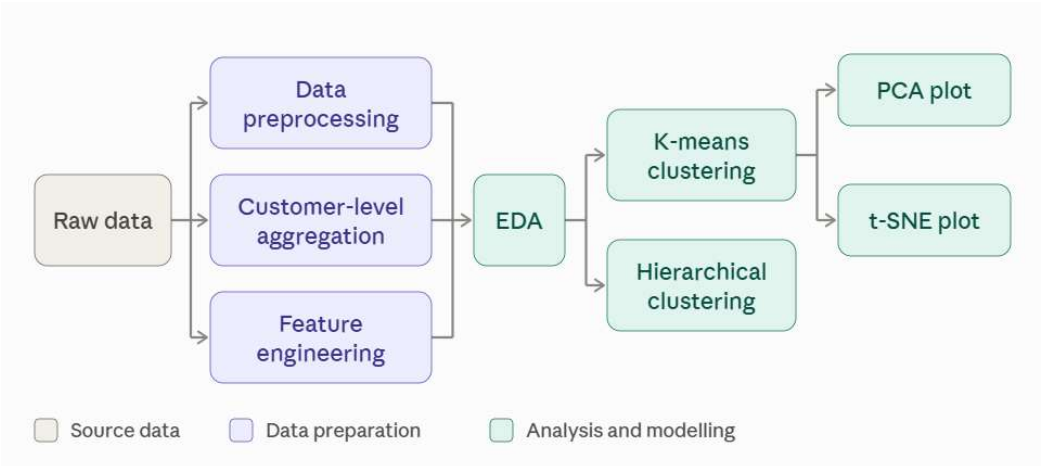


Figure 1: Overview of workflow

Results and Discussion

EDA

The dataset contained 747,953 orders and 68,300 customers taken between 2012-2016. Descriptive statistics of the selected features is presented (table 1). All features showed right skewness except for customer age showing symmetrical distribution. Outliers are visualised by box plots (fig.2) and identified by the IQR method accounting for 3.6%, 4.9%, 5.0%, 4.2% of Frequency, Recency, CLV, and Average_Unit_Cost respectively. No outliers were identified for customer age.

	Frequency	**Recency**	**CLV**	**Average_Unit_Cost**	**Customer age**
count	68300	68300	68300	68300	68300
mean	11.0	367.7	181.5	78.9	43.1
std	8.8	414.5	112.1	38.0	17.5
min	1.0	0.0	0.6	0.5	19.0
25%	5.0	45.0	120.0	57.6	28.0
50%	9.0	190.0	162.0	73.5	43.0
75%	15.0	539.3	215.1	92.5	58.0
max	101	1824	4089.1	1463.5	73.0

Table. 1 Summary statistics of selected features

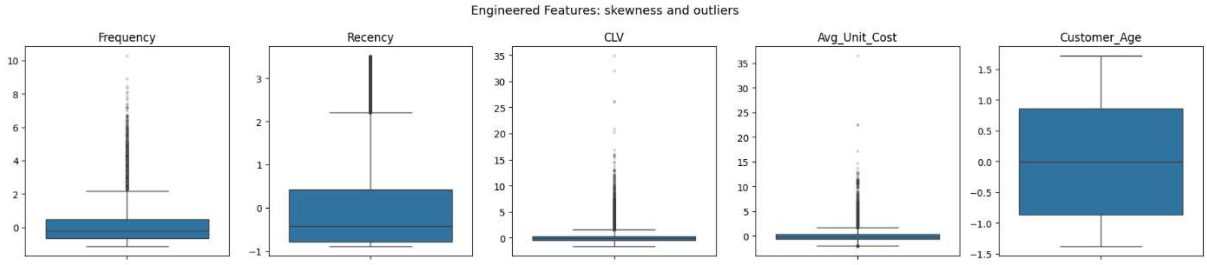


Figure. 2 Box plots of features

A strong correlation is observed between CLV and Average_Unit_Cost (0.73), suggesting revenue per order is driven by expensive the items. A weak/moderate correlation is observed between (-0.43) Recency and Frequency, suggesting that frequent buyers tend to be active (fig.3). Other remaining features show low pairwise correlation. The scatter plots of the features visualise these relationships (fig.4) showing no separation of groups.

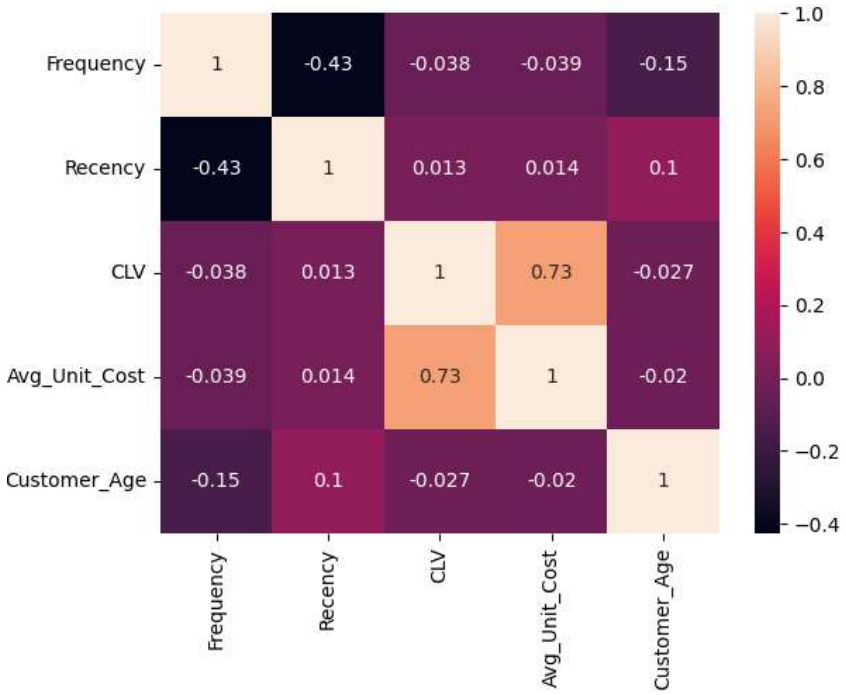


Figure. 3 Feature correlation heatmap

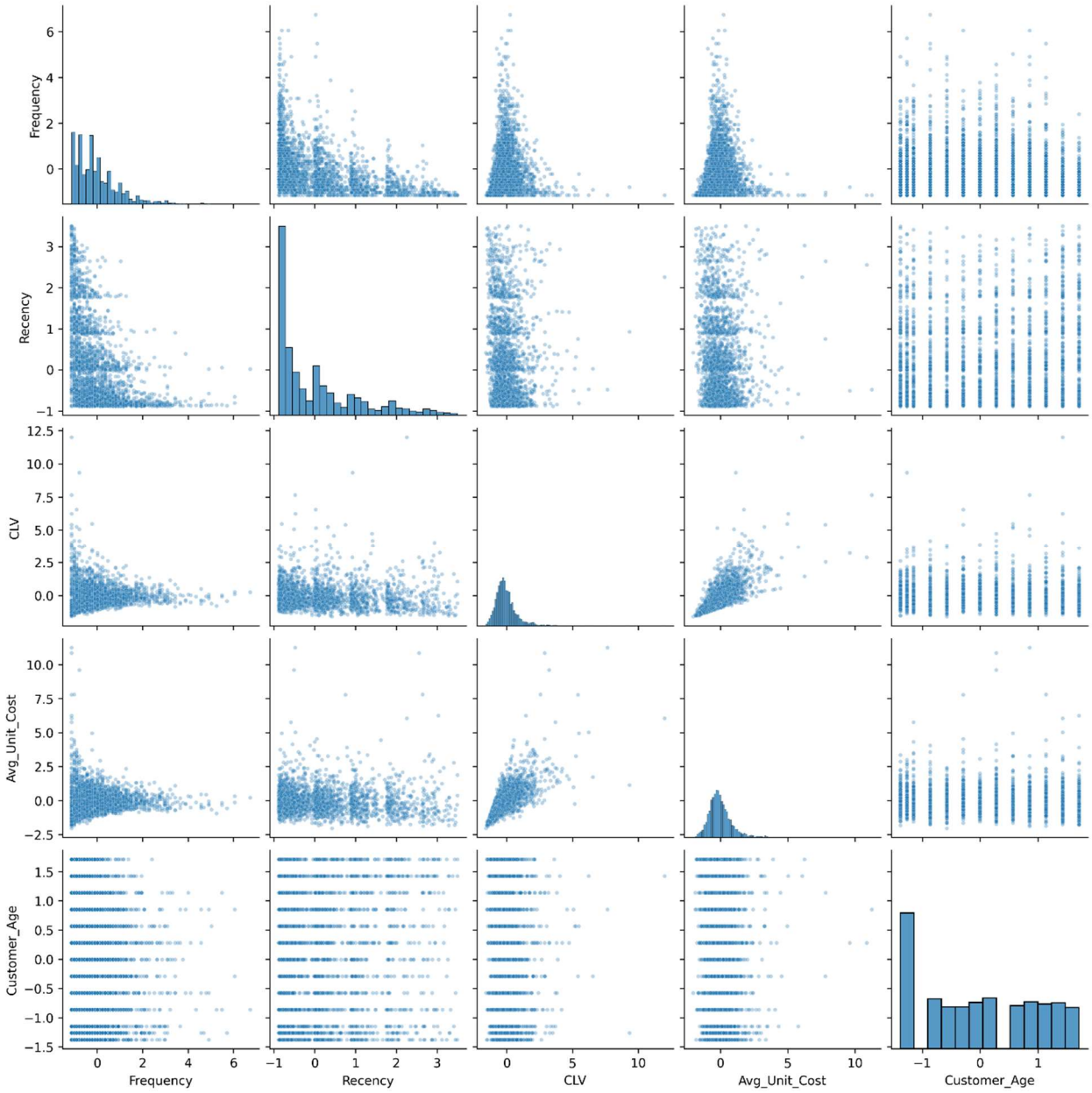


Figure 4 Feature scatter plot

Clustering with ML models

The elbow point was identified at k=6 for selection (fig.5) and the chosen silhouette score was 0.243 that agreed for six clusters. The silhouette analysis suggests possible overlaps with neighbouring clusters implying that clusters may not be very well separated (fig.6).

Hierarchical clustering was performed on several linkage methods (fig.7). Ward was considered the most appropriate method based on cut-off point and sensible cluster size populations.

Cluster profile was observed looking at the feature mean values taken from customer-level dataset, and inferred feature summary statistics (table.4).

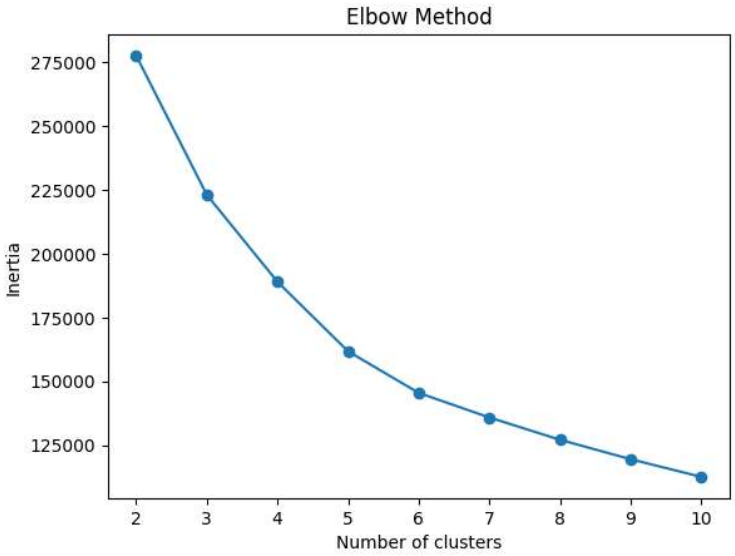


Figure 5. Elbow score

K	Silhouette Score
2	0.258
3	0.232
4	0.229
5	0.242
6	0.243
7	0.239
8	0.223
9	0.207
10	0.212

Table.3 Silhouette score

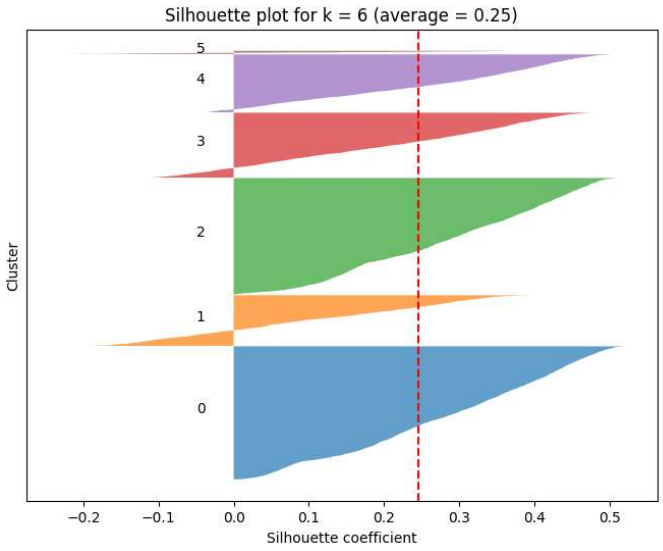


Figure. 6 Silhouette analysis for K-means cluster

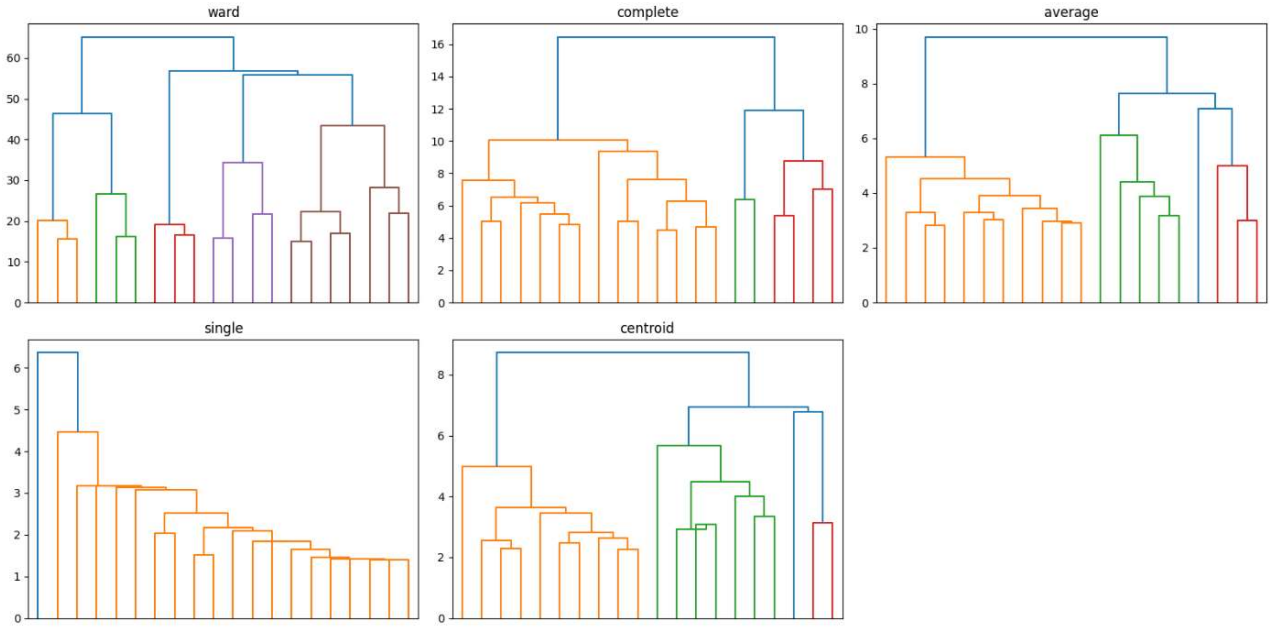


Figure 7. Dendrogram of selected features

Cluster	Customers	Frequency	Recency	CLV	Avg unit cost	Age	Profile
0	182208	9.2	224	150	69	61	Moderate spenders, old (27%)
1	8717	7.0	417	322	128	42	Infrequent high spenders (13%)
2	21248	9.9	229	154	70	28	Moderate spenders, young (27%)
3	9779	27.3	107	178	78	39	Highly frequent, average spenders (14%)
4	785	2.5	736	750	255	46	Occasional high spenders (1%)
5	9563	4.2	1140	131	61	48	Occasional low spender (14%)

Table .4 Cluster profile of feature mean values

Visualising customer segments and dimensionality reduction

Cluster Boxplots

The boxplots confirm the behavioural patterns described for the cluster profiles (fig.8, table.4). The boxplots for CLV and Average_Unit_Cost profile ae very similar showing the correlation between these two features.

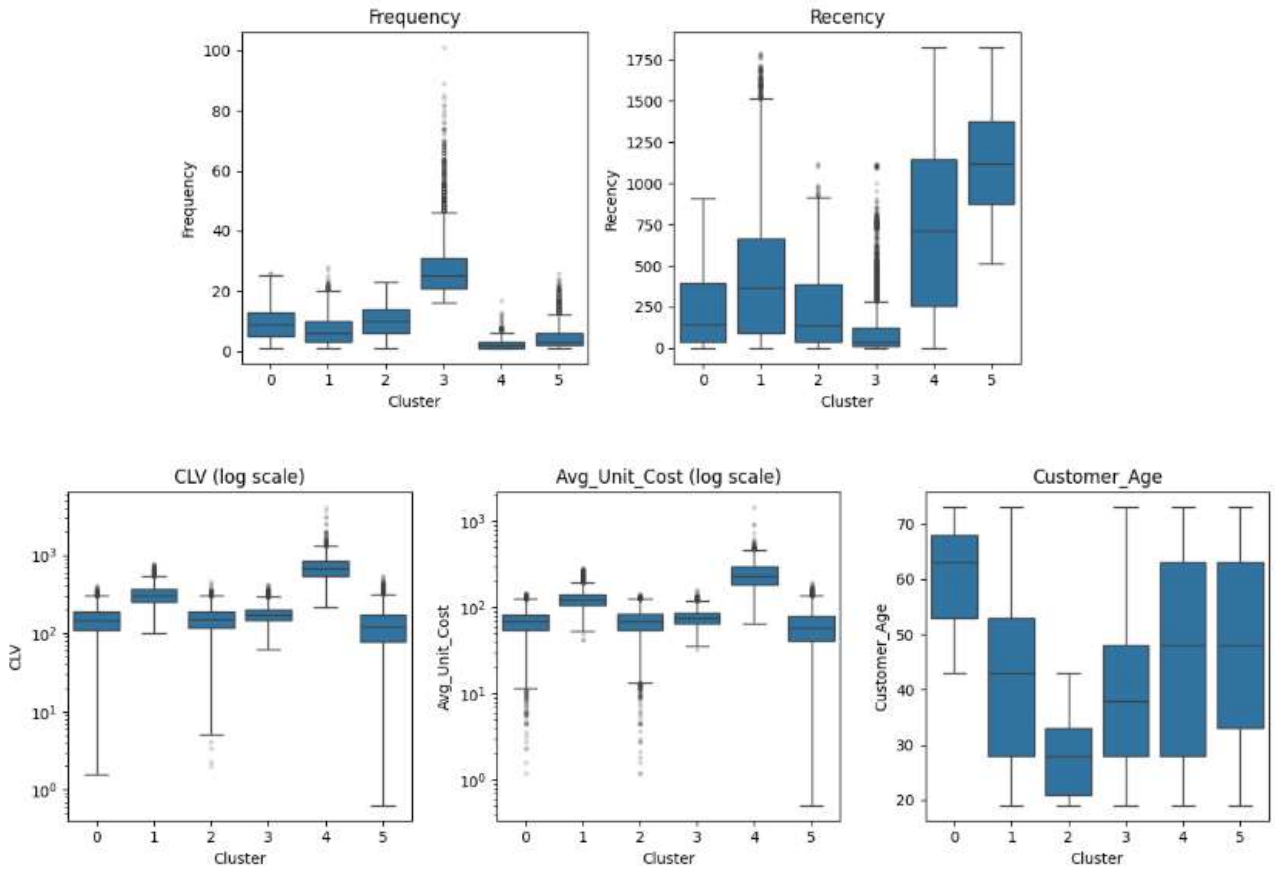


Figure. 8 Cluster Box plots

PCA 2D visualisation

The PCA plot shows the K-mean cluster separation but not into isolated groups. PC1 and PC2 together hold 65% of explained variation, so about a third is not shown here. Observing the loadings, PC1 is dominated by CLV, and Average_Unit_Cost, so moving right means higher-value customers. The extreme sparse points on PC1 likely represent the occasional high spenders. PC2 is dominated by low recency and high frequency, above 0 representing active customers. The moderate customers (clusters 0 and 2) differ by age (table.4) which explains the mixing of these

two groups. PC3 holds 19% of explained variance that is dominated by age, collectively giving 84% (table.5).

	PC1	PC2	PC3	PC4	PC5
Frequency	-0.13	0.66	0.18	0.72	0.00
Recency	0.10	-0.65	-0.31	0.69	0.00
CLV	0.70	0.11	0.04	0.02	0.71
Avg Unit Cost	0.70	0.11	0.05	0.02	-0.71
Customer Age	-0.00	-0.35	0.93	0.09	0.01
Explained variance	0.35	0.30	0.19	0.11	0.05

Table 5. Feature loading and explained variance

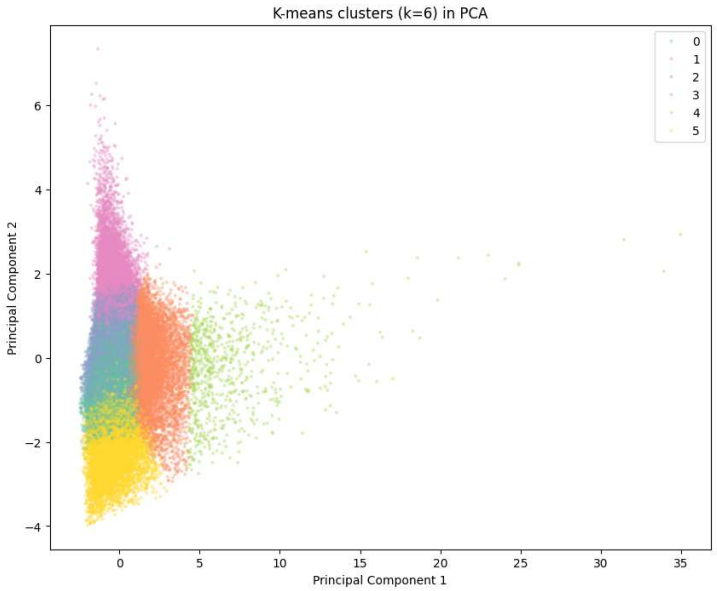


Figure 9. Cluster labelled PCA visual plot

t-SNE 2D visualisation

The t-SNE plot (fig.10) show the clusters but not as separate isolated groups with some mixing observed. (fig.6). The visualisation of the cluster arrangement is more locally coherent as clusters 0 and 2 show clear separation that is not shown in PCA.

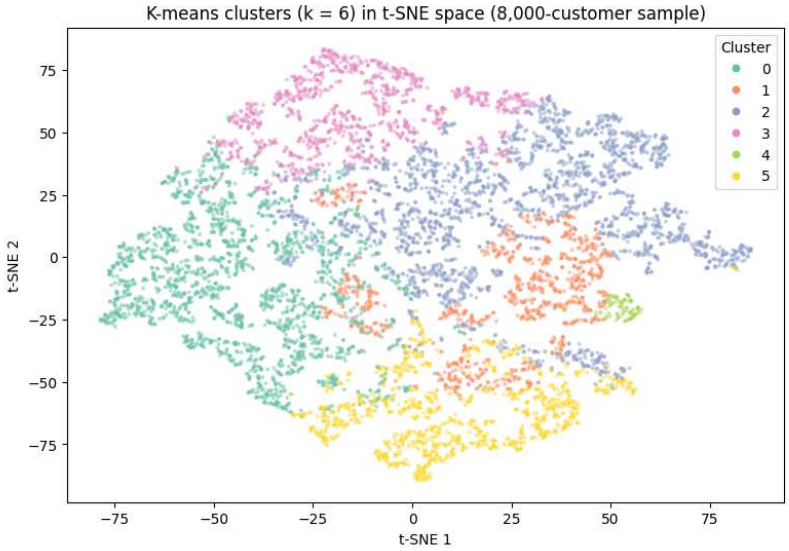


Figure 10. Cluster labelled t-SNE plot

Conclusion

The K-means (k=6) clustering model segmented 68,300 customers into distinct interpretable behavioural profiles that can be considered for targeted marketing strategies. The choice of k was supported by the elbow at 6 clusters, silhouette score of 0.24, and a ward dendrogram consistent with six branches. The interpretability of the clusters was supported by the cluster distribution in the feature box plots, while PCA and t-SNE showed segments representing the clusters.

The clusters, although defined, were not isolated separate groups suggesting that customer behaviour varies continuously. The soft elbow and the low silhouette scores implied overlapping clusters. K-means is sensitive to skewed features, and extreme values may contribute to this because distance-based measures are affected. This can pose a challenge for customer clustering as extreme values of features may represent valuable customer groups that you may not want to remove. This is demonstrated by the small cluster of about 785 customers (1%) having the highest CLV and unit cost, and it exists because those extreme values were kept.

Furthermore, CLV and Average_Unit_Cost are strongly correlated (r =0.73), so spending level may have carried more weight in the clustering profiling than other features. Further exploration could investigate the robustness of the clusters by removing one of the correlated features and investigating other features that may affect customer behaviour.